

# Meta-Learning for Crisis-Aware Portfolio Allocation: A Jump Model Approach

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**Abstract**—Financial crises pose significant challenges for portfolio allocation models due to their rarity, unique characteristics, and the need for rapid adaptation. Traditional machine learning approaches struggle during market disruptions because of data scarcity and distribution shifts. We present a novel application of Model-Agnostic Meta-Learning (MAML) combined with statistical jump detection for crisis-aware portfolio allocation. Our approach uses the Lee-Mykland jump model to detect regime changes and constructs meta-learning tasks from historical market periods, enabling rapid adaptation to emerging crises with limited data. Using 32 financial indicators derived from 35 years of S&P 500 and VIX data spanning seven major crises, our model achieves a test loss of 0.0251 (15.8% RMSE) and demonstrates strong out-of-distribution generalization on the COVID-19 crisis with only 1.87% prediction bias, representing a 91% improvement in bias reduction compared to baseline approaches. Portfolio backtesting validates practical effectiveness with a Sharpe ratio of 0.47, outperforming buy-and-hold (0.36) and traditional 60/40 allocation (0.32). Our results demonstrate that pattern diversity in training tasks is more critical than model complexity for small-data meta-learning regimes.

**Index Terms**—meta-learning, MAML, portfolio allocation, crisis prediction, regime detection, jump models, financial machine learning

## I. INTRODUCTION

Financial crises represent critical challenges for portfolio management systems. Market disruptions such as the 2008 financial crisis, 2020 COVID-19 pandemic, and 2022 Ukraine war create rapid shifts in asset behavior that traditional models fail to capture effectively. Three fundamental problems emerge: (1) data scarcity, with only 5-10 major crises occurring over 30 years; (2) distribution shift, where each crisis exhibits unique characteristics; and (3) adaptation speed, requiring models to adjust quickly as crises unfold.

Traditional machine learning approaches require substantial training data and struggle with out-of-distribution scenarios. Regime-switching models like Hidden Markov Models (HMMs) rely on predefined states and often exhibit lag in detecting transitions. Deep learning methods demand extensive retraining when market conditions change. These limitations motivate our investigation of meta-learning for crisis-aware portfolio allocation.

Meta-learning, or "learning to learn," offers a solution by training models to adapt quickly to new tasks with minimal

data. Model-Agnostic Meta-Learning (MAML) [1] has shown success in few-shot learning across computer vision and natural language processing. We apply MAML to financial crisis prediction, treating each historical market regime as a distinct task. Our model learns patterns across multiple crises, enabling rapid adaptation when new disruptions emerge.

Our key contributions are:

- A novel application of MAML to crisis-aware portfolio allocation using jump model regime detection
- Demonstration that pattern diversity in training tasks exceeds model complexity in importance for small-data meta-learning
- Strong out-of-distribution generalization validated on COVID-19 crisis (1.87% bias)
- Empirical validation through portfolio backtesting achieving 0.47 Sharpe ratio
- Competitive performance with 0.0251 test loss (15.8% RMSE) on our 35-year dataset

## II. RELATED WORK

### A. Regime Detection in Financial Markets

Regime detection methods aim to identify distinct market states. HMMs represent the classical approach, modeling market dynamics as transitions between hidden states [16]. However, HMMs suffer from lag and require careful threshold calibration. The Volatility Index (VIX) serves as a popular fear indicator but spikes reactively after crises begin rather than predictively.

Jump models provide an alternative framework based on statistical detection of discontinuous price movements. Lee and Mykland [12] developed a nonparametric test using bipower variation for robust volatility estimation. This approach detects actual market movements rather than sentiment, reducing lag compared to VIX-based methods.

### B. Meta-Learning

MAML [1] optimizes for rapid adaptation by training a model initialization that enables fast fine-tuning on new tasks. The algorithm maintains gradient flow through the adaptation process using higher-order derivatives. Applications span computer vision [41], robotics, and natural language processing.

Financial applications of meta-learning remain limited. Recent work explores few-shot learning for stock prediction [9] and portfolio optimization [7], but focuses on individual assets rather than crisis regimes. Our work applies MAML specifically to crisis-aware portfolio allocation with jump-based regime detection.

### C. Crisis Prediction and Portfolio Allocation

Traditional crisis prediction relies on early warning indicators such as credit spreads, yield curves, and volatility measures [27]. Machine learning approaches typically require retraining for new crises and lack fast adaptation mechanisms, motivating our meta-learning approach.

Portfolio optimization has incorporated machine learning techniques [18] and reinforcement learning [17], but adapting to sudden regime changes remains challenging. Our approach addresses this gap through meta-learning.

## III. METHODOLOGY

### A. Data and Features

We utilize 35 years of daily S&P 500 and VIX data (1990-2025, 8,970 trading days) from Yahoo Finance. Our feature engineering generates 32 financial indicators across five categories:

**Price Features (6):** S&P 500 log returns, moving averages (5-day, 20-day), 20-day volatility, 10-day momentum, and high-low range.

**Technical Indicators (3):** 14-day Relative Strength Index (RSI), 14-day Average True Range (ATR), and volume percent change.

**VIX Features (6):** VIX log returns, VIX level, VIX spike indicator, and binary thresholds for VIX above 20, 30, and 40.

**Volume Features (1):** Volume log returns for liquidity analysis.

**Imputed Features (16):** Forward-filled versions of all base features to handle missing data, ensuring continuous time series without gaps.

All features undergo standardization using zero-mean, unit-variance scaling to ensure stable gradient flow during training.

### B. Jump-Based Regime Detection

Our regime classification builds on the Lee-Mykland jump detection framework [12]. For each day  $t$ , we compute the test statistic:

$$L_t = \frac{r_t - \mu}{\sigma_t} \quad (1)$$

where  $r_t$  represents the log return,  $\mu$  denotes the local return mean, and  $\sigma_t$  indicates bipower variation providing robust volatility estimation:

$$BV_t = \sum_{i=1}^{n-1} |r_i| \times |r_{i-1}| \times \frac{\pi}{2} \quad (2)$$

We detect jumps when  $|L_t|$  exceeds a threshold derived from the inverse cumulative distribution function with 1% significance level. Jump classification includes binary detection, magnitude, and direction.

Using rolling 20-day windows, we compute jump frequency and average jump size to assign regime labels:

- **JR0 (Calm):** Jump frequency  $< 10\%$  and average size  $< 2\%$
- **JR1 (Moderate):**  $10\% \leq \text{frequency} < 20\%$  or size  $< 3\%$
- **JR2 (Crisis):** Frequency  $\geq 20\%$  or size  $\geq 3\%$

This approach captures seven major crises including the 1998 Asian crisis, 2002 WorldCom collapse, 2008 financial crisis, 2009 recovery, 2011 debt downgrade, 2020 COVID-19 pandemic, and 2025 market volatility, achieving superior coverage compared to VIX-only methods.

### C. Task Generation

Meta-learning requires multiple training tasks. We segment the time series at regime change points, growing segments while maintaining dominant regime purity above 60%. Tasks span 20-90 days, ensuring sufficient data for support-query splits while capturing crisis diversity.

Each task contains:

- **Support set:** 20 samples for adaptation
- **Query set:** 10 samples for evaluation
- **Metadata:** Regime label, purity, jump statistics

Our final dataset comprises 69 tasks: 29 JR0 (calm), 15 JR1 (moderate), and 7 JR2 (crisis). The temporal split ensures no data leakage: 35 training tasks (including 3 JR2), 8 validation tasks (2 JR2), and 26 test tasks (2 JR2 including COVID-19 as out-of-distribution holdout).

### D. MAML Training

We implement MAML using the learn2learn library [6], following the algorithm specification in [1]. Our architecture consists of a 2-layer multilayer perceptron with 64 hidden units predicting next-day log returns.

The training process alternates between inner and outer loops:

**Inner Loop (Adaptation):** For each task, we clone the meta-model and perform 5 gradient descent steps on the support set:

$$\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\text{support}}(\theta) \quad (3)$$

where  $\alpha = 0.01$  represents the inner learning rate.

**Outer Loop (Meta-Update):** We evaluate adapted parameters on query sets and compute the meta-loss:

$$\mathcal{L}_{\text{meta}} = \frac{1}{|B|} \sum_{i \in B} \mathcal{L}_{\text{query}}(\theta'_i) \quad (4)$$

where  $B$  denotes the task batch. The meta-optimizer (Adam with learning rate 0.001) updates the base parameters to minimize query loss across all tasks:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}_{\text{meta}} \quad (5)$$

This procedure enables the model to learn an initialization that adapts quickly to new market regimes with minimal data.

#### IV. EXPERIMENTS AND RESULTS

##### A. Training Configuration

We train for 50 epochs with batch size 4 (tasks per batch) on Tesla T4 GPU hardware. Training completes in 12 minutes, demonstrating computational efficiency. We apply gradient clipping (max norm 5.0) to both inner and outer loops for stability.

##### B. Overall Performance

Table I presents our main results. The model achieves 0.0251 test loss (15.8% RMSE) across all regimes, with strong performance in calm markets (15.3% RMSE) and moderate conditions (14.7% RMSE). Crisis prediction reaches 23.6% RMSE, representing competitive performance on our dataset.

TABLE I  
TEST PERFORMANCE BY REGIME

| Regime         | Loss (MSE)    | RMSE (%)    | Tasks     |
|----------------|---------------|-------------|-----------|
| JR0 (Calm)     | 0.0234        | 15.3        | 18        |
| JR1 (Moderate) | 0.0215        | 14.7        | 6         |
| JR2 (Crisis)   | 0.0557        | 23.6        | 2         |
| <b>Overall</b> | <b>0.0251</b> | <b>15.8</b> | <b>26</b> |

##### C. Crisis Task Analysis

Table II details performance on all seven crisis tasks. Our model demonstrates strong generalization on COVID-19 (test set) with only 1.87% positive bias despite never observing similar V-shaped recoveries during training. The 2011 debt crisis shows exceptional accuracy (1.48% bias), while the 2008 financial crisis remains challenging with 24.8% positive bias reflecting extreme severity.

TABLE II  
CRISIS TASK PERFORMANCE ANALYSIS

| Event          | Period                           | Split    | Loss          | Bias          |
|----------------|----------------------------------|----------|---------------|---------------|
| 1998 Asian     | Aug 31 - Sep 30 (21d)            | Train    | 0.0046        | +2.12%        |
| 2002 WorldCom  | Jul 23 - Aug 22 (22d)            | Train    | 0.0073        | +2.32%        |
| 2008 Financial | Sep 18, 2008 - Jan 6, 2009 (75d) | Train    | 0.0735        | +24.8%        |
| 2009 Recovery  | Feb 24 - Apr 7, 2009 (30d)       | Val      | 0.0151        | +5.63%        |
| 2011 Debt      | Aug 9 - Sep 8 (21d)              | Val      | 0.0084        | +1.48%        |
| 2020 COVID     | Mar 4 - Apr 24 (36d)             | Test     | 0.0111        | +1.87%        |
| 2025 Recent    | Apr 7 - May 6 (20d)              | Test     | 0.0358        | +4.44%        |
| <b>Average</b> | <b>-</b>                         | <b>-</b> | <b>0.0223</b> | <b>+4.83%</b> |

##### D. Portfolio Backtesting

We validate practical effectiveness through portfolio backtesting on the same 35-year period. Using MAML predictions for regime-aware allocation (60% equity in calm markets, 50% in moderate volatility, 30% in crises), our strategy achieves:

- Sharpe Ratio: 0.47 (vs. Buy-and-Hold 0.36, 60/40 0.32, 40/60 0.24)
- Cumulative Return: 19.9x (vs. Buy-and-Hold 18.7x)
- Annualized Return: 8.9% with 14.8% volatility
- Maximum Drawdown: -50.8% (similar to Buy-and-Hold -56.8%)

These results demonstrate that meta-learned crisis predictions translate to superior risk-adjusted portfolio returns.

##### E. Ablation Studies

We conducted extensive ablation studies to isolate key factors:

**Task Threshold Impact:** Lowering the minimum task duration from 30 to 20 days increased crisis task count from 5 to 7, improving test loss by 35.9% and crisis loss by 69.7%. This demonstrates that pattern diversity dominates model complexity.

**Gradient Flow:** Manual MAML implementation using `copy.deepcopy()` failed catastrophically, while the `learn2learn` library with proper higher-order gradients succeeded immediately. This validates the critical importance of maintaining computational graphs through adaptation.

**Jump Model vs. VIX:** Our jump-based regime detection captured the 2022 Ukraine war and 2002 WorldCom crisis missed by VIX-only approaches, contributing to improved coverage.

**Bias Evolution:** Comparison with baseline approach (30-day threshold, 5 crisis tasks) shows significant improvement: COVID-19 bias reduced from -22.5% (defensive) to +1.87% (nearly neutral), representing 91% bias reduction.

##### F. Model Characteristics

Our MAML approach demonstrates fast adaptation capability, requiring only 5 gradient steps on 20 support samples to adapt to new market regimes. This enables real-time deployment when crises emerge, with adaptation completing in approximately 100 milliseconds on standard hardware. The meta-learned initialization encodes transferable patterns across diverse crisis types, enabling strong out-of-distribution generalization as demonstrated by the COVID-19 results.

#### V. DISCUSSION

Our results validate meta-learning for crisis-aware portfolio allocation. The strong COVID-19 generalization (1.87% bias) despite out-of-distribution characteristics demonstrates that MAML learns transferable crisis patterns rather than memorizing historical events.

The critical role of pattern diversity emerged as our key finding. Adding two diverse crises (2002 sharp correction and 2025 volatility) to training yielded greater improvement than any architectural modification. This insight guides future work: expanding to international markets to capture 15-20 crisis tasks should further enhance generalization.

The jump model framework proved superior to VIX-based detection by capturing actual price movements rather than

sentiment. Jump frequency and magnitude provide leading indicators that reduce lag compared to reactive fear measures.

Limitations include the 2008 outlier (24.8% bias) indicating extreme events remain challenging, and the requirement for 7+ diverse crisis tasks limiting applicability to short historical datasets. The portfolio backtest assumes zero transaction costs and perfect execution, which may overestimate real-world performance.

Future work should explore: (1) international market expansion for diverse crisis patterns, (2) hierarchical MAML for crisis subtypes, (3) incorporation of transaction costs in portfolio optimization, and (4) real-time deployment and monitoring systems.

## VI. CONCLUSION

We presented a novel application of meta-learning combining MAML with jump model regime detection for crisis-aware portfolio allocation. Our approach achieves competitive crisis prediction accuracy (23.6% RMSE on crisis regimes) and strong out-of-distribution generalization (1.87% COVID-19 bias) while adapting rapidly to new crises with only 20 samples and 5 gradient steps.

The validation that pattern diversity exceeds model complexity in importance for small-data meta-learning provides actionable guidance for practitioners: invest in diverse training scenarios rather than architectural sophistication when working with limited crisis data.

Portfolio backtesting demonstrates practical effectiveness with 0.47 Sharpe ratio, outperforming traditional strategies on our 35-year dataset. The combination of statistical jump detection and meta-learning represents a principled approach to the fundamental challenges of crisis prediction: data scarcity, distribution shift, and adaptation speed.

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